Electromagnetic multipole moments

Separate perturbative and twist contributions with lattice normalization

3 October 2026

What does the contact contribution mean?

In these tables, **contact** is the retained electromagnetic Wilson-line response of the local condensate part of the two-point OPE. A Wilson line connects the separated charged fields and supplies their gauge phase. Expanding that line to first order in the external photon field gives an additional contribution to the three-point correlator.

With the straight-link prescription used in this calculation, the stored response is

$$\Pi^W_{\mu\nu} = -Q_H \epsilon^\rho \int_0^1 d\beta \left. \frac{\partial C^{\rm cond}_{0,\mu\nu}(P;p,p')}{\partial P^\rho} \right|_{P=p+\beta q}, \qquad Q_H = e_Q - e_s.$$

Here $C_0^{\rm cond}$ is the physical two-point condensate tensor; the minus sign follows our stored convention $\Pi = -C$. Only the internal momentum P is differentiated. External current momenta p and $p' = p + q$ remain fixed. Replacing ϵ by q turns the integral into the difference of condensate tensors at the two endpoints, giving the required electromagnetic Ward response of this block.

This contribution is kept separate from the photon distribution-amplitude sectors. The perturbative photon-on-quark response is already included in the perturbative sector and is not added again here. The link response survives the double Borel transform, so the word "contact" does not mean that it is discarded from the moments. Ward-response checks support this prescription; they do not uniquely determine all transverse contributions.

Why is it zero for Bs? Every retained link kernel is proportional to Q_H. For these charged Ds currents, $Q_H = 1$; for the neutral Bs currents, $Q_H = 0$. The Bs contact entries therefore vanish exactly in this prescription. The contact block is independent of χ and is unchanged by reversing twist 2.

Inputs and reading the tables

The supplied lattice normalization is $f_{\perp\gamma,s}(1\,\text{GeV}) = -55.1$ MeV. The $+\chi$ label uses this value; $-\chi$ uses $+55.1$ MeV and flips only twist 2. The condensate remains $\langle \bar{s}s \rangle = 0.8(-0.240\,\text{GeV})^3$ in all other terms. Chi enters only twist 2, not the two-point function or contact block.

Other inputs: $\zeta_1 = 0.4$, $\zeta_2 = 31/110$, $\kappa = 0.2$, $\kappa^+ = 0$, $f_{3\gamma} = -0.004$ GeV^2, $\omega_V = 3.8$, $\omega_A = -2.1$, $m_0^2 = 0.8$ GeV^2. The DA scale is 1 GeV. Nominal angles are 26.6° (Ds) and 38.5° (Bs).

Nominal mixing angles

Accepted-window counts in state order: 37, 25, 57, 17.

Each entry is the median of the indicated contribution. Twist-2 columns give both signs explicitly; the other columns apply to both signs.

Magnetic moment: μ/μ_N

State	Perturbative	Twist 2 (+χ)	Twist 2 (-χ)	Twist 3	Twist 4	Contact
$D_{s1}(2460)^+$	0.780010	-0.568491	0.568491	-0.094003	-0.069647	-0.015260
$D_{s1}(2536)^+$	0.638959	0.051564	-0.051564	-0.337387	-0.156162	0.097233
$B_{s1}(5750)$	0.133431	-0.314355	0.314355	-0.017663	0.017337	0.000000
$B_{s1}(5830)$	0.480480	0.023798	-0.023798	-0.329725	0.206843	0.000000

Quadrupole form factor: $F_D - Q_H$

State	Perturbative	Twist 2 (+χ)	Twist 2 (-χ)	Twist 3	Twist 4	Contact
$D_{s1}(2460)^+$	-2.053195	-2.643512	2.643512	0.590327	-0.154038	0.701903
$D_{s1}(2536)^+$	-2.216542	5.355061	-5.355061	1.791186	-0.072485	-1.020116
$B_{s1}(5750)$	-0.147135	-1.432447	1.432447	-0.387724	0.266620	0.000000
$B_{s1}(5830)$	-5.728128	18.334977	-18.334977	8.001665	-8.241479	0.000000

Grouping and normalization

Perturbative: heavy- and strange-line photon emission. **Twist 2:** leading photon DA with lattice normalization. **Twist 3:** two-particle twist 3 plus three-particle V and A. **Twist 4:** two-particle twist 4 plus S, S-tilde, T1-T4 and electromagnetic S-gamma and T4-gamma. **Contact:** Wilson-line condensate response described on page 1.

The magnetic contributions use F_M from $q_\mu \epsilon_\nu$ and $\mu/\mu_N = (m_p/m_H)F_M$. The quadrupole columns give the individual terms in $F_D - Q_H = -F_M + G_3$ using the clean G_3 projection. To obtain F_D, add $Q_H = 1$ for Ds and $Q_H = 0$ for Bs at each window. The dimensional quadrupole moment is $\mathcal{D}/e = F_D(\hbar c)^2/m_H^2$ in fm^2.

Medians of the parts need not sum to the median total. Additivity holds at each window: the largest residual is 3.91×10^{-14}. The six printed decimals are retained for reproducibility, not an estimate of physical precision. Accepted-window spreads are not a full theoretical uncertainty; the results use the central LO/twist and continuum prescription.

90 degrees: refitted pure-current windows

Accepted-window counts in state order: 55, 79, 64, 30. The lighter state uses J_A and the heavier state uses J_B.

Each entry is the median of the indicated contribution. Twist-2 columns give both signs explicitly; the other columns apply to both signs.

Magnetic moment: μ/μ_N

State	Perturbative	Twist 2 (+χ)	Twist 2 (-χ)	Twist 3	Twist 4	Contact
$D_{s1}(2460)^+$	0.839351	-0.476423	0.476423	0.026448	-0.057200	-0.002975
$D_{s1}(2536)^+$	0.707351	-0.470276	0.470276	-0.188986	-0.089902	0.000371
$B_{s1}(5750)$	0.160582	-0.294854	0.294854	-0.019698	0.022629	0.000000
$B_{s1}(5830)$	0.114196	-0.260607	0.260607	-0.020594	0.012360	0.000000

Quadrupole form factor: $F_D - Q_H$

State	Perturbative	Twist 2 (+χ)	Twist 2 (-χ)	Twist 3	Twist 4	Contact
$D_{s1}(2460)^+$	-0.283352	1.248852	-1.248852	-1.329886	0.005127	0.026241
$D_{s1}(2536)^+$	-2.984153	-3.452141	3.452141	1.946491	-0.256859	0.786012
$B_{s1}(5750)$	1.032532	1.807016	-1.807016	-0.828522	-0.207140	0.000000
$B_{s1}(5830)$	-1.204771	-2.942746	2.942746	0.277191	0.343996	0.000000

Grouping and normalization

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Provenance

All entries are read from our signed summary CSV frozen in Git commit `54e6958`; no counterpart numerical outputs were used. The source definitions and straight-link audit are in our recalculation report and `blind_definition_scan_audit/AUDIT.md`. The original CSV SHA-256 hashes are unchanged. A separate manifest covers this handout and its PDF.